

Quantitative Methods for Business and Economics

Lecture 6 - Single Variable Calculus & Optimization

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What is Calculus?

- Calculus is the mathematical study of change.
- It has two main branches:
 - **Differential calculus**: studies the rate at which quantities change.
 - **Integral calculus**: studies the accumulation of quantities.
- Calculus is used in many fields, including physics, engineering, economics, and social sciences.

Differential Single Variable Calculus

↳ one thing changing
at time..

$$f(x) = x^2$$

↳ one variable

- Single variable calculus deals with functions of one variable.

- Functions of one variable take a single input and produce a single output. This is usually denoted as:

↳ real → real numbers

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x, y) = x^2 + y^2$$

↳ two

- Differential calculus studies the rate at which functions change.
Hence, rates of change are the main focus of this branch of calculus.

↳ one variable
at a time.

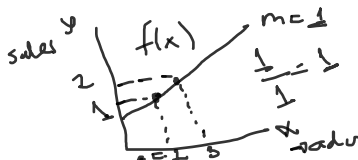
Rates of Change

Δ delta = change x, y, z

- Remember variations Δx and Δy ? Rates of change for functions are not too different from these.
- The rate of change of a function $f(x)$ at a point $x = a$ represents how much the function changes as x changes by a small amount Δx around a .
- The concept of slope is crucial for understanding rates of change, as it is the ratio of the change in y to the change in x .
- The slope is not constant for all functions. It can change depending on the point at which it is calculated.

$$y = mx + b$$

$\hookrightarrow$ slope = change !



Slopes and Tangent Lines

- The slope of the tangent line to the graph of $f(x)$ at $x = a$ is the rate of change of $f(x)$ at $x = a$.

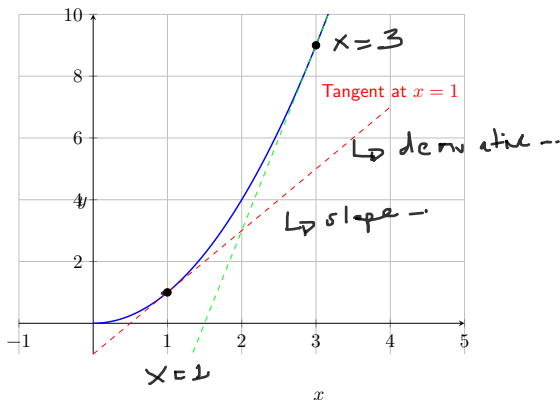


Figure 1: Curve $f(x) = x^2$ and its tangent lines at $x = 1$ and $x = 3$

Derivatives

- Because rates of change are not constant, it makes sense to define a function that gives the rate of change of another function at any point.

- This function is called the **derivative** (or the first derivative). It is

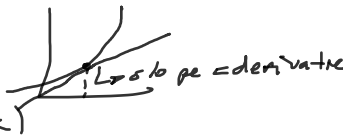
denoted as $f'(x)$ or $\frac{df}{dx}$ → not a fraction

- The derivative of a function $f(x)$ at a point $x = a$ is the slope of the tangent line to the graph of $f(x)$ at $x = a$. ↳ any number

- The derivative of a function $f(x)$ is itself a function that gives the rate of change of $f(x)$ at any point.

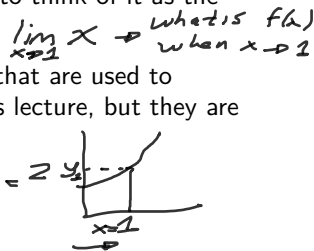
slope of the tangent

↳ touches $f(x)$



A brief mention of limits

- The derivative of a function $f(x)$ at a point $x = a$ is defined as the limit of the average rate of change of $f(x)$ as x approaches a .
- Limits are a fundamental concept in calculus. They are used to define derivatives, integrals, and many other concepts in calculus.
- A limit is an operator that gives the value a function approaches as the input approaches a certain value. It helps to think of it as the value the function gets closer and closer to.
- There are many rules and properties of limits that are used to calculate them. We will not cover them in this lecture, but they are crucial for a deeper understanding of calculus.



The Derivative of a Function

- The derivative of a function $f(x)$ at a point $x = a$ is defined as:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

→ formal concept of derivative

- This definition gives the slope of the tangent line to the graph of $f(x)$ at $x = a$.
- The derivative of a function $f(x)$ is itself a function that gives the rate of change of $f(x)$ at any point.
- The derivative of a function $f(x)$ is denoted as $f'(x)$ or $\frac{df}{dx}$.

Derivatives without limits

- Thankfully, we don't really need limits to calculate derivatives for most functions.
- The rules of differentiation allow us to calculate the derivative of a function without using limits, but rather relatively simple algebraic manipulations.

Rules of Differentiation

$$f'(x) \rightarrow f(x) = x \Rightarrow 1x^0 = 1 \quad x^3 \rightarrow 3x^{3-1} = 3x^2$$

1 Power rule: If $f(x) = x^n$, then $f'(x) = nx^{n-1}$. $\rightarrow x^2 \rightarrow 2x^{2-1} = 2x$

2 Sum rule: If $f(x) = g(x) + h(x)$, then $f'(x) = g'(x) + h'(x)$.

3 Product rule: If $f(x) = g(x)h(x)$, then $f'(x) = g'(x)h(x) + g(x)h'(x)$.

$$\hookrightarrow x^3 + x^2$$

4 Quotient rule: If $f(x) = \frac{g(x)}{h(x)}$, then

$$3x^2 + 2x$$

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{(h(x))^2}$$

$$\frac{x^5 + x^4}{5x^4 + 4x^3}$$

5 Chain rule: If $f(x) = g(h(x))$, then $f'(x) = g'(h(x))h'(x)$.

$$f(x) = 3x^2 \cdot 4x^5$$

$$f'(x) = (6x \cdot 4x^5) + 3x^2 \cdot 20x^4$$

$$\left. \begin{array}{l} 2x^6 \\ 12x^5 \end{array} \right\}$$

More properties of derivatives

■ The derivative of a constant is zero: If $f(x) = c$, then $f'(x) = 0$.

■ The derivative of the identity function is one: If $f(x) = x$, then $f'(x) = 1$.

■ For the exponential function $f(x) = a^x$, the derivative is $f'(x) = a^x \ln(a)$.

$f(x) = 3^{100} \rightarrow f'(x) = 3^{100} \ln 3$
 $f(x) = 55^x \rightarrow f'(x) = 55^x \ln 55$

■ For the natural logarithm function $f(x) = \ln(x)$, the derivative is $f'(x) = \frac{1}{x}$.

■ For the trigonometric functions, the derivatives are as follows:

- $\sin(x)$: $\cos(x)$
- $\cos(x)$: $-\sin(x)$
- $\tan(x)$: $\sec^2(x)$
- $\cot(x)$: $-\csc^2(x)$
- $\sec(x)$: $\sec(x) \tan(x)$
- $\csc(x)$: $-\csc(x) \cot(x)$

$f(x) = \sin x \rightarrow f'(x) = \cos x$
 $f(x) = \cos x \rightarrow f'(x) = -\sin x$

Example: Derivative of a power function

- Let's calculate the derivative of the function $f(x) = x^2$.
- Using the power rule, we have:

$$f'(x) = 2x^{2-1} = 2x$$

- Therefore, the derivative of $f(x) = x^2$ is $f'(x) = 2x$.

Example: Derivative of a sum

- Let's calculate the derivative of the function $f(x) = x^2 + 3x$.
- Using the sum rule, we have:

$$f'(x) = (x^2)' + (3x)' = 2x + 3$$

- Therefore, the derivative of $f(x) = x^2 + 3x$ is $f'(x) = 2x + 3$.

Example: Derivative of a product

- Let's calculate the derivative of the function $f(x) = x^2 \cdot \sqrt{x}$.
- Using the product rule and the power rule, we have:

$$f'(x) = (x^2)' \cdot \sqrt{x} + x^2 \cdot (\sqrt{x})'$$

$$f'(x) = 2x \cdot \sqrt{x} + x^2 \cdot \frac{1}{2\sqrt{x}}$$

$$f'(x) = 2x\sqrt{x} + \frac{x^2}{2\sqrt{x}}$$

Example: Applying the chain rule

- Let's calculate the derivative of the function $f(x) = (x^2 + 1)^3$.
- The chain rule involves two functions: the outer function $g(x) = x^3$ and the inner function $h(x) = x^2 + 1$.
- Using the chain rule, we have:

$$f'(x) = 3(x^2 + 1)^2 \cdot (2x)$$

$$f'(x) = 6x(x^2 + 1)^2$$

Applications: the concept of marginality

- In many fields, the derivative of a function has a special meaning.
- The concept of **margin** is crucial in economics and business, denoting the **additional** or **incremental** change in a quantity given a small change in another quantity.
- The derivative of a function represents the marginal change in the function with respect to the input.
- Marginal products, marginal costs, marginal revenues, and marginal utilities are all examples of marginal quantities that are calculated using derivatives.

Example: marginal profit

Suppose a car dealership's profit function is

$$\Pi(q) = 200q - q^2$$

where q is the number of cars sold.

Handwritten notes:

- $\Pi(q)$ is labeled with an arrow pointing to " π ".
- $f(x)$ is written below $\Pi(q)$.
- An arrow points from q to a circled "quantity".

Handwritten note:

profits are related to n° we sell.

Example: marginal profit

What is the marginal profit? This means we need to calculate the derivative of profit with respect to the number of cars sold:

$$\boxed{\frac{d\Pi}{dq}} = 200 - 2q$$

$$200q - q^2$$
$$200 - 2q$$

- This derivative is the marginal profit function.
- It tells us approximately how much profit changes when the dealership sells one more car.
- For example, at $q = 50$, $\rightarrow 50^{\text{th}}$ car

$$MP(50) = 200 - 2(50) = 100 \checkmark$$

+100 extra \$ of profit

- So near 50 cars sold, one additional car raises profit by about \$100.

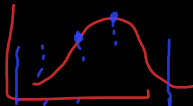
Single variable optimization

- Optimization is a key concept in the social sciences and many other fields.
- Optimization involves finding the maximum or minimum value of a function, sometimes subject to constraints.
- Calculus, and specifically derivatives, are crucial tools for optimization.
- In single-variable optimization, we are interested in finding the maximum or minimum value of a function of one variable.
- For now, we will focus on finding the maximum or minimum value of a function without constraints.

$\max \pi \dots$

$\min C \dots$

Critical points



↳ both maximum, minimum, inflection point

- Critical points are points at which the derivative of a function is zero or undefined.
- Critical points are important because they can be local maxima, local minima, or neither.
 - Local maxima are points where the function reaches a maximum value in a small neighborhood.
 - Local minima are points where the function reaches a minimum value in a small neighborhood.
 - Some critical points are neither a maximum nor a minimum.
- To find the critical points of a function, we need to find the values of x for which $f'(x) = 0$ or $f'(x)$ is undefined.

$$\boxed{f'(x) = 0} \rightarrow \text{critical points}$$

Example: Finding critical points

Consider the function $f(x) = x^3 - 3x^2 + 2x$.

To find the critical points of $f(x)$, we need to find the values of x for which $f'(x) = 0$ or $f'(x)$ is undefined. *→ does not matter*

First, we need to calculate the derivative of $f(x)$:

$$f'(x) = \underline{3x^2 - 6x + 2}$$

Now, we need to find the values of x for which $f'(x) = 0$:

$$3x^2 - 6x + 2 = 0$$
$$ax^2 + bx + c = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

(1) + (2) -

Example: Finding critical points

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4 \cdot 3 \cdot 2}}{2 \cdot 3}$$

$$x = \frac{6 \pm \sqrt{36 - 24}}{6}$$

$$x = \frac{6 \pm \sqrt{12}}{6}$$

$$x = \frac{6 \pm 2\sqrt{3}}{6}$$

$$x = 1 \pm \frac{\sqrt{3}}{3}$$

$$x_1 = 1 + \frac{\sqrt{3}}{3}$$

$$x_2 = 1 - \frac{\sqrt{3}}{3}$$

Example: Finding critical points

- Therefore, the critical points of the function $f(x) = x^3 - 3x^2 + 2x$ are $x = 1 + \frac{\sqrt{3}}{3}$ and $x = 1 - \frac{\sqrt{3}}{3}$.
- To determine whether these points are local maxima, local minima, or saddle points, we need to analyze the behavior of the function around these points.
- There are several ways to do this:
 - Use the second derivative test (we will cover this later)
 - Look at the graph of the function around the critical points.

Example: Finding critical points

- Let's first look at the graph of $f(x) = x^3 - 3x^2 + 2x$ around the critical points.

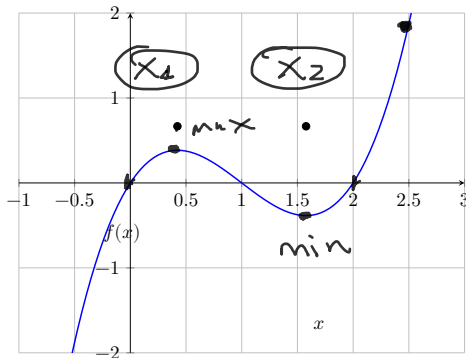


Figure 2: Graph of the function $f(x) = x^3 - 3x^2 + 2x$

Example: Finding critical points

- From the graph, we can see that the function has a local maximum at

$$x = 1 + \frac{\sqrt{3}}{3}$$

and a local minimum at

$$x = 1 - \frac{\sqrt{3}}{3}.$$

- Therefore, the critical points of the function $f(x) = x^3 - 3x^2 + 2x$ are $x = 1 + \frac{\sqrt{3}}{3}$ and $x = 1 - \frac{\sqrt{3}}{3}$.

Maximum and minimum values with the second derivative test

- Once we have found the critical points of a function, we can determine whether they are local maxima, local minima, or saddle points.
- To determine whether a critical point is a local maximum or a local minimum, we can use the second derivative test apart from looking at the graph.
- The second derivative is calculated as the derivative of the first derivative of the original function $f(x)$. (+)
- The second derivative test states that if $f''(x) > 0$ at a critical point $x = a$, then $f(a)$ is a local minimum. If $f''(x) < 0$ at a critical point $x = a$, then $f(a)$ is a local maximum. (-)

Example: Using the second derivative test

Consider the function $f(x) = x^3 - 3x^2 + 2x$.

We have already found the critical points of this function:

$$\text{and } x = 1 - \frac{\sqrt{3}}{3}.$$

$$x = 1 + \frac{\sqrt{3}}{3}$$

Now, we need to calculate the second derivative of $f(x)$:

$$f'(x) = 3x^2 - 6x + 2$$

↗ 0 ↗ second derivative

$$f''(x) = 6x - 6$$

$f''(x)$

Example: Using the second derivative test

$$f'(x) = 6x - 6$$

Now, we need to evaluate $f''(x)$ at the critical points:

$$(1) \quad f''\left(1 + \frac{\sqrt{3}}{3}\right) = 6\left(1 + \frac{\sqrt{3}}{3}\right) - 6 = 6 + 2\sqrt{3} - 6 = 2\sqrt{3} > 0 \quad (+) \rightarrow \text{minimum}$$

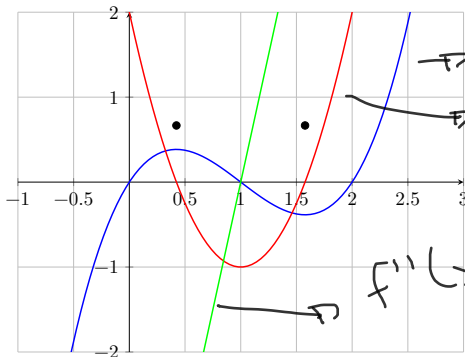
$$(2) \quad f''\left(1 - \frac{\sqrt{3}}{3}\right) = 6\left(1 - \frac{\sqrt{3}}{3}\right) - 6 = 6 - 2\sqrt{3} - 6 = -2\sqrt{3} < 0 \quad (-) \rightarrow \text{maximum}$$

Example: Using the second derivative test

- Therefore, the critical point $x = 1 + \frac{\sqrt{3}}{3}$ is a local minimum, and the critical point $x = 1 - \frac{\sqrt{3}}{3}$ is a local maximum.
- This means that the function $f(x) = x^3 - 3x^2 + 2x$ has a local minimum at $x = 1 + \frac{\sqrt{3}}{3}$ and a local maximum at $x = 1 - \frac{\sqrt{3}}{3}$.

Visualizing the function and derivatives

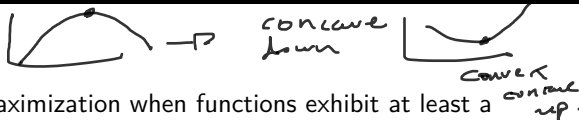
- Let's visualize the function $f(x) = x^3 - 3x^2 + 2x$ and its first and second derivatives.
 - *Blue* represents the function $f(x)$.
 - *Red* represents the first derivative $f'(x)$.
 - *Green* represents the second derivative $f''(x)$.



Local and global maxima and minima

- As you've noticed, the function above had both a local maximum and a local minimum.
- We call these points local since the function goes up and down around them, so in principle they are not the maximum or minimum values of the function.
- We then say that the function has no **global** maximum or minimum.
- Some functions may have local and global maxima or minima, while others may have only local maxima or minima. Others may have neither.
- The existence of global maxima and minima is a more complex problem that involves analyzing the behavior of the function at the boundaries of its domain, and we will not cover it in this lecture.

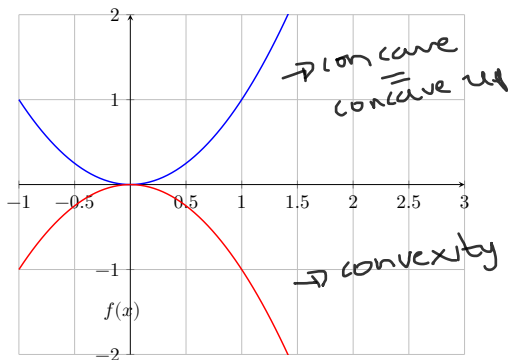
Maximization and minimization



- We are interested in maximization when functions exhibit at least a local maximum, and in minimization when they exhibit at least a local minimum.
- Whether a function has maxima or minima is a question of interest in many fields, and it depends on the shape of the function and the behavior of its derivatives.
- We often talk about concavity and convexity of functions to determine whether they have maxima or minima.
- As a rough rule, if a function is concave up, it tends to have a minimum, and if it is concave down, it tends to have a maximum.

Concavity and convexity - graphical intuition

- A function is **concave up** if it curves upwards, like a smiley face.
- A function is **concave down** if it curves downwards, like a frown.
This is convexity.



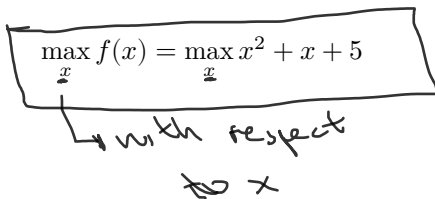
Optimization notation

- A real world example of optimization is maximizing or minimizing a function that represents a quantity of interest.
 - For example, a government may want to maximize public welfare given a certain budget.
 - A firm may want to maximize profit or minimize cost.
 - An investor may want to choose the scale of an investment to maximize returns net of cost.
- In all these cases, optimization involves finding the maximum or minimum value of a function. This is done by finding the critical points of the function and analyzing their behavior.
- The notation of these optimization problems is usually as follows:

Maximum
operates $\rightarrow$ $\max f(x)$ or $\min f(x)$ *minimum*

Optimization notation

- The objective function $f(x)$ represents the quantity we want to maximize or minimize.
- The objective variable x represents the variable that we can change to maximize or minimize the function.
- We often say that we maximize or minimize a function **with respect to** (w.r.t.) the objective variable. In notation, this is written as:


$$\max_{\underline{x}} f(x) = \max_{\underline{x}} x^2 + x + 5$$

with respect to x